

A Canonical Common-Weight All- m Type-II Estimate and the Complementary Signed High-Order Frontier

Draft V public research snapshot: Gate 1A, Gate 1B, provenance, and open closure
ledger

Research draft – not a claim of Full Type II or twin-prime closure

August 26, 2026

Abstract

This note records the current public-facing state of a parity-sensitive Type-II proof programme. Part I isolates a canonical common-weight direct operator. For a prime row $r \asymp R$, shift k , and $m \asymp M$, the exact determinant identity $m' A_e(t) - m B_e(t) = 2k$ leads, after source-preserving Poisson–Bruhat transport, to a moving-row family $c_r(X) = B_X F_X(r/R)$. Band-limited prime participation, Huxley’s short-interval prime theorem, a fixed-state codegree bound, and one outer square root yield the required $R^{-1/4+o(1)}$ contraction on the parameter polytope, conditional on the explicit source interfaces stated below. The projective branch is grouped by the actual ratio class Z/L , not by the product ZL ; this matches the machine-checked collision relation $Z_1 L_2 = Z_2 L_1$.

Part II records the complementary Gate-1B frontier. The low/medium conductor range is controlled. In the high-conductor H7 branch, the source prime is forced into the primitive conductor; the short inducing-cofactor Möbius sign cancels exactly against the induced Gauss factor. Exact prime-character collapse followed by hybrid Poisson summation in the long h -slot recovers the entire previously missing fixed power Q/Y^4 and reduces the problem to a natural-scale dual determinant. The remaining H7 task is an arbitrary-log saving for a centred seven-defect determinant variance. H8, H9, the same- q Gram residual, a source-specific cross-modulus residual, the historical/canonical zero-mode lock, and Full Type-II reassembly remain open.

The status firewall is deliberate: finite algebra and exponent ledgers that are machine-checked are distinguished from external analytic inputs, source/provenance pins, and genuinely open estimates. No claim of Full Type II reassembly, a Hardy–Littlewood prime-pair asymptotic, or twin-prime infinitude is made.

Contents

1	Introduction	2
1.1	External analytic input	3
1.2	Scope	3
2	Parameters and notation	3
3	The canonical physical source	4
3.1	Rows and affine forms	4
3.2	Common physical smoothing	4
3.3	Source provenance and derived row dependence	5
4	Fourier normalization and determinant shell	5

5	Source-preserving Poisson–Bruhat transport	6
6	Band-limited prime participation	6
6.1	Projective versus non-projective Gram pairs	8
7	The unique outer square root	8
8	Projective reassembly: ratio classes, not product classes	9
9	Recombination error and provenance boundary	9
10	Main theorem	10
11	Scope separation: Gate 1A versus source routing	10
12	Formal verification and provenance	11
12.1	Machine-checked finite/source skeleton	11
12.2	What Lean does not prove	11
12.3	Research status versus formal status	12
13	Relation to Kloosterman-fraction technology	12
14	What remains outside this theorem	12
15	Conclusion	13
A	Formal theorem map and scope ledger	13
B	Canonical source versus arbitrary edge dependence	13
C	Complementary programme boundary: Gate 1B, current authoritative snapshot	14
C.1	Low/medium conductor: closed at fixed-power scale	14
C.2	Exact source split in high conductor	15
C.3	Prime-character collapse: exact	15
C.4	Hybrid h -Poisson and the dual determinant	16
C.5	The fixed-power wall is gone; the log-saving wall remains	17
C.6	What remains after H7	17
C.7	Technology map and backend audit	18
C.8	Gate-1B closure DAG	18
D	Publication-facing theorem/status map	19

Public status firewall (Draft V). Gate 1A finite/compiler algebra: PASS. Gate 1A end-to-end analytic theorem: CONDITIONAL on the explicit source/provenance hypotheses in Theorems 5.1, 6.2, 8.1 and 9.1. Gate 1B: OPEN. Full Type-II reassembly: OPEN. Ford–Maynard downstream transfer is therefore not activated. Twin-prime infinitude is not claimed.
--

1 Introduction

A recurring obstruction in parity-breaking sieve arguments is that strong fixed-slice estimates need not survive coherent averaging over the true source family. The correct object is therefore the source-restricted family operator, not a collection of unrelated local norms.

This note isolates a canonical direct Type-II operator and proves an all- m mean-square bound for it. Four structural ingredients drive the proof:

- (i) exact source and determinant-2 row geometry;
- (ii) source-preserving Fourier/Poisson transport to a smooth moving- r family;
- (iii) band-limited prime participation, converting smoothness in r into a moving-family contraction;
- (iv) an exact projective crossed-convolution identity for the zero-curvature branch.

The downstream motivation is the Ford–Maynard prime-producing sieve framework, where Type I and Type II information for an actual non-negative sequence is used simultaneously [1]. Their work also shows that a substantial Type II range is genuinely necessary for non-trivial lower bounds. We do not perform the full downstream reassembly here.

1.1 External analytic input

The participation step needs a prime number theorem in intervals of length $R^{3/4}$. Huxley’s classical short-interval prime number theorem gives, uniformly for $x \asymp R$,

$$\pi(x+H) - \pi(x) = (1 + o(1)) \frac{H}{\log x} \quad (H \geq x^{7/12+\varepsilon}),$$

for every fixed $\varepsilon > 0$ [2]. Since $3/4 > 7/12$, the plateaux used below are longer than the range required by Huxley’s theorem, so no modern strengthening is needed. We also use the classical Bernstein inequality

$$\|P'\|_\infty \leq N \|P\|_\infty$$

for trigonometric polynomials of degree at most N [3].

1.2 Scope

The theorem proved below concerns the canonical *common-weight* direct source. In the authoritative source, the physical smoothing datum is one weight W_D ; row dependence that appears later is derived deterministically from this common datum through Fourier, reciprocity and Poisson transforms. An abstract interface allowing arbitrary independent edge-dependent weights is strictly more general and belongs to an upstream source-adaptation problem. Likewise, a global packet census and Full Type II reassembly are not part of the theorem proved here. This distinction is also machine-checked in the formal development: the Gate-1A direct closure certificate contains no global packet-dictionary or edge-dependent-D2 field.

2 Parameters and notation

Let $X \rightarrow \infty$. Set

$$M = X^{1/3}, \quad R = X^a, \quad L = X^b,$$

and

$$H = X^{a+2b-2/3}, \quad K = \frac{M}{R}, \quad U = \frac{L}{H}, \quad D = \frac{L^2}{H}.$$

We repeatedly use

$$RK = M, \quad DH = L^2, \quad M^2 H = RL^2. \tag{1}$$

The three controlling vertices are

$$V_1 = \left(\frac{5}{18}, \frac{1}{3} \right), \quad V_2 = \left(\frac{5}{18}, \frac{25}{72} \right), \quad V_3 = \left(\frac{7}{24}, \frac{1}{3} \right).$$

For a prime ℓ , define

$$\rho_\ell(n) := \mathbf{1}_{\ell|n} - \frac{1}{\ell}. \quad (2)$$

Let b_p, d_q be 1-bounded coefficient sequences on primes $p, q \asymp L$. All harmless divisor and logarithmic losses are absorbed in $X^{o(1)}$.

3 The canonical physical source

3.1 Rows and affine forms

A canonical row is $e = (r, k, m)$ with

$$r \asymp R \text{ prime}, \quad 1 \leq |k| \leq K, \quad m \asymp M, \quad m' := m + kr \asymp M,$$

together with the generic unit/full-conductor conditions needed by the reciprocal phases. Let $\mathcal{E}^\sharp$ be this row family.

Choose w_0 modulo r such that

$$mw_0 \equiv -2 \pmod{r}.$$

Since $m' = m + kr \equiv m \pmod{r}$, the same choice also satisfies $m'w_0 \equiv -2 \pmod{r}$. Hence

$$\alpha := \frac{mw_0 + 2}{r}, \quad \beta := \frac{m'w_0 + 2}{r}$$

are integers. Define

$$A_e(t) := \alpha + mt, \quad B_e(t) := \beta + m't.$$

Lemma 3.1 (Determinant-2 row identity). *For every canonical row and every integer t ,*

$$m'A_e(t) - mB_e(t) = 2k.$$

Proof. The t -terms cancel and

$$m'\alpha - m\beta = \frac{m'(mw_0 + 2) - m(m'w_0 + 2)}{r} = \frac{2(m' - m)}{r} = 2k.$$

□

3.2 Common physical smoothing

Fix $W \in C_c^\infty(\mathbb{R})$ and write $W_D(t) = W(t/D)$. The key point is that this weight is common to all rows. Define

$$P_e(t) = \sum_{p \asymp L} b_p \rho_p(A_e(t)), \quad Q_e(t) = \sum_{q \asymp L} d_q \rho_q(B_e(t)),$$

and

$$C_e(b, d) := \sum_{t \in \mathbb{Z}} W_D(t) P_e(t) Q_e(t). \quad (3)$$

Remark 3.2. An arbitrary family $W_{D,e}$ is not part of the canonical theorem. Such weights arise only in a more general source-adaptation interface.

3.3 Source provenance and derived row dependence

The common-weight assertion is stronger than a notational convention. After Fourier transformation the source coefficient factors as

$$\omega_{m,p,q}(h) = \underbrace{\frac{H}{pq} e\left(-\frac{h\alpha}{pqm}\right)}_{\text{derived row-dependent scalar}} \underbrace{\widehat{W}_D\left(\frac{h}{pq}\right)}_{\text{one common physical weight}}, \quad (4)$$

where the argument $h/(pq)$ has no row index. Thus the later dependence on m, r, p, q is *derived dependence*; it is not freedom to choose an independent physical weight on each row. This distinction is load-bearing: an arbitrary edge-dependent family would be a strictly stronger source class and is deliberately excluded from the canonical direct theorem.

4 Fourier normalization and determinant shell

The normalized pre-square source is

$$\tilde{C}_{r,k,m} = \sum_{p \asymp L} b_p \sum_{\substack{q \asymp L \\ q \neq p}} d_q \sum_h \omega_{m,p,q}(h) e_{mq}\left(2hk\overline{p(m+kr)}\right), \quad (5)$$

where

$$\omega_{m,p,q}(h) = \frac{H}{pq} \widehat{W}_D\left(\frac{h}{pq}\right) e\left(-\frac{h\alpha}{pqm}\right). \quad (6)$$

The source norms satisfy

$$\sum_h |\omega_{m,p,q}(h)|^2 \ll H, \quad \sum_h |\omega_{m,p,q}(h)| \ll H.$$

Writing $m' = m + rk$, additive reciprocity separates the phase:

$$e_{mq}(2hk\overline{pm'}) = e_{pr}(-2h\overline{qm}) e\left(\frac{2h}{prqm}\right) e_q(-2h\overline{prmm'}). \quad (7)$$

Thus the exact H -scale smooth profile is

$$\beta_{m,r;p,q}(h/H) = \frac{H}{pq} \widehat{W}_D(h/pq) e\left(-\frac{h\alpha}{pqm} + \frac{2h}{prqm}\right).$$

The h -Poisson/Gauss completion gives the determinant shell

$$ms - prn = 2. \quad (8)$$

Choose $\kappa_p(m) \in [0, pr)$ with $m\kappa_p(m) \equiv 2 \pmod{pr}$ and set

$$n_0(m, p, r) = \frac{m\kappa_p(m) - 2}{pr}.$$

Then every solution is parameterized by

$$s_j = \kappa_p(m) + jpr, \quad n_j = n_0 + jm.$$

The quotient amplitude is

$$\mathcal{W}_{m,r;p,q}(j) := \widehat{\beta}_{m,r;p,q}\left(\frac{H(n_0 + jm)}{qm}\right), \quad (9)$$

and, for $|j| \ll U = L/H$,

$$\sum_{|j| \ll U} |\mathcal{W}_{m,r;p,q}(j)|^2 \ll UX^{o(1)}. \quad (10)$$

5 Source-preserving Poisson–Bruhat transport

The following source-specific transport statement is isolated explicitly, because it is the bridge between the authoritative physical source and the finite BPP compiler. Its algebraic normalization is elementary, while the uniform smooth-envelope statement is an analytic/source input.

Hypothesis 5.1 (One-sided source transport). *After quotient-cell recombination and corrected two-dimensional Poisson–Bruhat transport, each effective state X has scalar moving-row amplitude*

$$c_r(X) = B_X F_X(r/R), \quad (11)$$

where B_X is independent of r , F_X is supported in a fixed compact interval, and for every fixed J ,

$$\sup_X \sup_t |\partial_t^J F_X(t)| \ll_J X^{o(1)}.$$

The transport is one-sided: it preserves the required source upper norm with only $X^{o(1)}$ nuclear cost; no two-sided Hilbert–Schmidt equivalence is assumed. In addition, for every fixed A_0 the transported source admits a finite retained-state set $\mathcal{X}_{\text{ret}}$ and a discarded remainder whose contribution to the required Gate energy is $O_{A_0}(X^{-A_0})$ times the natural scale. On every retained state the Fourier truncation below is small relative to the envelope maximum, uniformly in the state.

Status note 5.2 (Pending source pin). The finite Poisson–Bruhat algebra and normalization are banked, but the full source-preserving smooth-envelope statement, including the negligible-state disposal just stated, remains a source/analytic input. This is a **PENDING** closure item, not a Lean theorem.

Lemma 5.3 (Poisson–Bruhat budget identity). *The scale identities imply*

$$\boxed{\frac{MK(MH)^2}{RL^4} = 1.} \quad (12)$$

Proof. By (1), $MK = L^2/H$. Therefore

$$\frac{MK(MH)^2}{RL^4} = \frac{(L^2/H)M^2H^2}{RL^4} = \frac{M^2H}{RL^2} = 1.$$

□

Remark 5.4. The amplitude is smooth in r but not constant. Hence a mere bound on the number of prime divisors of a fixed determinant does not itself give an R^{-1} weighted-family saving.

6 Band-limited prime participation

From now on X ranges over the retained state set $\mathcal{X}_{\text{ret}}$ provided by Theorem 5.1; the discarded states have already been absorbed in the $O_{A_0}(X^{-A_0})$ source remainder. Set $a_{r,X} = |c_r(X)|$ for a prime row $r \in [R, 2R]$, with the convention $a_{r,X} = 0$ when r is excluded for the state X . Define the continuous envelope

$$M_X := |B_X| \sup_{1 \leq t \leq 2} |F_X(t)|.$$

Fix a large A and set $N = R^{1/4}$. Uniform smoothness gives

$$F_X(t) = P_X(t) + O_A(R^{-A}), \quad (13)$$

where P_X is a trigonometric polynomial of degree at most N .

Lemma 6.1 (Bernstein plateau). *For every retained state X , there is an interval $I_X \subset [R, 2R]$ of length $\gg R^{3/4}$ on which*

$$|c_r(X)| \gg M_X.$$

Proof. Let t_X maximize $|P_X|$. Bernstein's inequality [3] gives $\|P'_X\|_\infty \leq N\|P_X\|_\infty$, so on an interval of t -length $\gg N^{-1}$ around t_X the polynomial retains a fixed proportion of its maximum. By (13), so does F_X . Multiplying the interval length by R gives $R/N = R^{3/4}$. $\square$

The only source-specific input needed to convert this continuous plateau into an admissible-prime plateau is the following fixed-state exclusion statement.

Hypothesis 6.2 (Fixed-state exclusion). *For each effective state X , every prime row excluded from the generic non-projective branch divides one of $X^{o(1)}$ fixed nonzero integers $N_{X,j}$ of polynomial height, $|N_{X,j}| \leq X^{O(1)}$.*

Lemma 6.3 (Admissible prime participation). *Assume Theorem 6.2. For every retained state X , the plateau I_X contains $R^{3/4-o(1)}$ admissible prime rows. Consequently, if*

$$A_X := \max_{\substack{r \sim R \\ r \text{ prime}}} a_{r,X}, \quad N_X := \frac{\sum_{\substack{r \sim R \\ r \text{ prime}}} a_{r,X}}{A_X},$$

then

$$A_X \asymp M_X, \quad N_X \gg R^{3/4-o(1)}. \quad (14)$$

Proof. By Theorem 6.1, I_X has length $\gg R^{3/4}$. Huxley's short-interval prime number theorem [2] applies because $3/4 > 7/12 + \varepsilon$ for a fixed sufficiently small $\varepsilon > 0$; hence I_X contains $R^{3/4-o(1)}$ primes. Each nonzero integer of polynomial height has $O(\log X)$ distinct prime divisors, and there are only $X^{o(1)}$ obstruction integers, so at most $X^{o(1)}$ primes are deleted. On all remaining plateau primes $a_{r,X} \gg M_X$, which gives both assertions in (14). $\square$

Let

$$\mathcal{R} := \{r \in [R, 2R] : r \text{ prime}\}, \quad S := |\mathcal{R}| = R^{1+o(1)},$$

and

$$\mathcal{T}_{\text{abs}} = \sum_{r \in \mathcal{R}} \left(\sum_X a_{r,X} \right)^2.$$

Lemma 6.4 (Participation-to-envelope). *If $N_X \geq P$ for every retained state X , then*

$$\left(\sum_X A_X \right)^2 \leq \frac{S}{P^2} \mathcal{T}_{\text{abs}}.$$

Proof. By Cauchy–Schwarz,

$$\left(\sum_X A_X N_X \right)^2 = \left(\sum_{r \in \mathcal{R}} \sum_X a_{r,X} \right)^2 \leq S \mathcal{T}_{\text{abs}}.$$

Since $N_X \geq P$, the claim follows. $\square$

6.1 Projective versus non-projective Gram pairs

The Poisson–Bruhat/projective coordinates attach to each effective state X a pair (Z_X, L_X) and the outer determinant

$$\Delta(X, Y) := Z_X L_Y - Z_Y L_X. \quad (15)$$

We partition the Gram pair set *before* using BPP:

$$\Delta(X, Y) \neq 0 \quad \text{or} \quad \Delta(X, Y) = 0.$$

The second class is the projective branch treated in Section 8. In particular,

$$\Delta(X, X) = 0,$$

so the state diagonal $X = Y$ is wholly contained in the projective branch. There is therefore no unsaved state-diagonal term hidden inside the BPP non-projective estimate.

For the non-projective branch, a fixed pair X, Y with $\Delta(X, Y) \neq 0$ can couple in row r only if r divides the fixed nonzero polynomial-height integer $\Delta(X, Y)$. Hence each such pair occurs in $X^{o(1)}$ rows and

$$\mathcal{E}_{\text{np}} \ll X^{o(1)} \left(\sum_X A_X \right)^2. \quad (16)$$

Combining with (14) yields:

Proposition 6.5 (BPP non-projective family-energy contraction).

$$\boxed{\mathcal{E}_{\text{np}} \ll R^{-1/2+o(1)} \mathcal{T}_{\text{abs}}.}$$

7 The unique outer square root

At the outer four-cycle level,

$$\|A_r\|_{\text{op}}^2 \leq \mathcal{T}_r,$$

so

$$\sum_{r \sim R} \|A_r\|_{\text{op}} \leq R^{1/2} \left(\sum_{r \sim R} \mathcal{T}_r \right)^{1/2}.$$

Hence the relative family-energy gain $R^{-1/2+o(1)}$ becomes $R^{-1/4+o(1)}$ at the Gate variance level. The required comparison is

$$MR^{-1/4} \leq H,$$

equivalently

$$\frac{5}{4}a + 2b - 1 \geq 0.$$

At the three controlling vertices the margins are

(a, b)	$\frac{5}{4}a + 2b - 1$
$(5/18, 1/3)$	$1/72$
$(5/18, 25/72)$	$1/24$
$(7/24, 1/3)$	$1/32$

and are all strictly positive.

8 Projective reassembly: ratio classes, not product classes

The complementary Gram branch is $\Delta(X, Y) = 0$, including the complete state diagonal $X = Y$. The collision relation is

$$Z_1 L_2 = Z_2 L_1,$$

so on the generic sector $L \neq 0$ the correct class map is the projective ratio

$$\kappa(Z, L) := Z/L \in \mathbb{Q}.$$

The product ZL is *not* the relevant class: equal products and equal projective ratios are different equivalence relations. This distinction is machine-checked by the hostile counterexample `Gate1A.Delta4.product_class_ne_ratio_class`.

Let $\mathcal{S}$ be the finite transported state support and write $v_{Z,L} = A_Z \otimes \overline{B_L}$. For each ratio class ξ define

$$C_\xi := \sum_{(Z,L) \in \mathcal{S}: \kappa(Z,L)=\xi} v_{Z,L}.$$

The finite class pushforward identity gives

$$\sum_{\xi} \|C_\xi\|^2 = \sum_{(Z_1,L_1),(Z_2,L_2) \in \mathcal{S}} \mathbf{1}_{Z_1 L_2 = Z_2 L_1} \operatorname{Re} \langle v_{Z_1,L_1}, v_{Z_2,L_2} \rangle. \quad (17)$$

This is the exact finite identity formalized as `Gate1A.Delta4.class_energy_identity` together with `ratioClass_eq_iff_cross`.

If each projective class contains at most τ states, then the banked finite multiplicity estimate gives

$$\sum_{\xi} \|C_\xi\|^2 \leq \tau \sum_{(Z,L) \in \mathcal{S}} \|A_Z\|^2 \|B_L\|^2 \leq \tau \left(\sum_Z \|A_Z\|^2 \right) \left(\sum_L \|B_L\|^2 \right). \quad (18)$$

On positive primitive representatives, vanishing projective defect is even rigid: the machine-checked theorem `Gate1A.SafeAlgebra.projective_zero_defect_rigidity` shows that $Z_1 L_2 = Z_2 L_1$ forces $(Z_1, L_1) = (Z_2, L_2)$. Primitivity is load-bearing.

Hypothesis 8.1 (Projective source routing). *The transported generic $\Delta = 0$ source is routed into positive primitive ratio-class packets with $\tau = X^{o(1)}$ class multiplicity; one-zero axes and non-primitive/rank-loss sectors are routed to separately controlled exceptional blocks at diagonal scale.*

Status note 8.2 (Projective status). The ratio-class identity, the collision equivalence, the finite multiplicity bound, and primitive rigidity are Lean-proved finite algebra. The physical source assertion that every relevant projective state satisfies the required primitive/axis routing is a **PENDING** source pin unless supplied by the upstream compiler. No product-fibre grouping is used in Draft V.

9 Recombination error and provenance boundary

Hypothesis 9.1 (Source recombination estimate). *The quotient-cell/source recombination error satisfies*

$$\mathcal{E}_{\text{rec}} \ll U^{-2+o(1)} M^2 L^4. \quad (19)$$

The exponent budget for this input is machine-checked. Since $U = L/H$ and $DH = L^2$,

$$\frac{U^{-2} M^2 L^4}{M H L^4} = \frac{M}{D}.$$

At V_1, V_2, V_3 the exponent margins are respectively

$$\frac{1}{18}, \quad \frac{1}{18}, \quad \frac{1}{24},$$

so (19) is more than sufficient once its literal source provenance is pinned.

Status note 9.2 (Recombination status). The U^{-2} arithmetic budget is Lean-proved; the literal analytic/source estimate (19) must be cited to the authoritative source theorem or retained as an explicit hypothesis. Draft V therefore marks this item **PENDING** rather than silently promoting budget arithmetic into a source estimate.

10 Main theorem

Theorem 10.1 (Canonical all- m common-weight Type-II estimate). *Let the parameters lie in the polytope generated by V_1, V_2, V_3 . Let $\mathcal{E}^\sharp$ be the generic unit/full-conductor all- m row family and let $C_e(b, d)$ be the canonical common-weight covariance (3). Assume Theorems 5.1, 6.2, 8.1 and 9.1. Then*

$$\sum_{e \in \mathcal{E}^\sharp} |C_e(b, d)|^2 \ll \frac{ML^4}{H} X^{o(1)}.$$

Equivalently, in the normalized Fourier convention,

$$\sum_{e \in \mathcal{E}^\sharp} |\tilde{C}_e(b, d)|^2 \ll MHL^4 X^{o(1)}.$$

Proof. Partition the transported Gram pairs by the outer determinant (15). The $\Delta \neq 0$ branch consists only of distinct states and is controlled by Theorem 6.5 followed by the unique outer square root; the parameter inequality is verified in the preceding section. The entire $\Delta = 0$ branch — including $X = Y$ — is the projective ratio-class energy (17) and is bounded at diagonal scale by (18) together with Theorem 8.1. The recombination error is acceptable by Theorem 9.1 and the U^{-2} exponent budget. These three classes are exhaustive and disjoint, so their sum proves the normalized estimate. Restoring the H^{-1} normalization on each amplitude yields the physical estimate. $\square$

Corollary 10.2 (Clean- P_3 packet). *Every clean- P_3 subfamily of $\mathcal{E}^\sharp$ satisfies the same bound.*

Proof. The energy is a sum of non-negative terms, so restriction to a subfamily only decreases it. $\square$

11 Scope separation: Gate 1A versus source routing

The theorem above is an analytic theorem for the canonical direct operator. It does not assert that every packet in a global sieve decomposition has already been identified with this operator.

Suppose an upstream source compiler gives an exact decomposition

$$\mathcal{S}_{\text{actual}} = \sum_{\lambda \in \Lambda_{\text{gen}}} a_\lambda D_\lambda U_\lambda \mathcal{S}_\lambda^\sharp + \mathcal{S}_{\text{exc}},$$

where U_λ is unitary/permutational and

$$\sum_{\lambda} |a_\lambda| \|D_\lambda\|_{\text{op}} \leq X^{o(1)}.$$

Then the main theorem controls the generic packets by Minkowski, provided each packet is truly an instance of the canonical operator. Proving the source identity, packet multiplicity, and exceptional routing is a separate Gate-0 compiler problem.

In particular, a common physical weight routes directly into the theorem. A fixed-dimensional smooth family generated deterministically from the common source can be decomposed into $X^{o(1)}$ common templates. By contrast, an arbitrary independently varying edge weight is a genuinely stronger object. In the formal development this more general datum is retained as an upstream Gate-0 source-adaptor obligation; it is not a premise of the canonical Gate-1A direct theorem.

This separation matters downstream. The Ford–Maynard framework applies to the actual non-negative sequence and its complete Type-I/Type-II information [1]; a convenient analytic model cannot replace an unverified source census.

12 Formal verification and provenance

The accompanying Lean bank deliberately separates finite mathematics, source provenance, and deep analytic input. This is useful here because a theorem compiler can be perfectly correct while its external analytic hypothesis is not itself formalized in Mathlib.

12.1 Machine-checked finite/source skeleton

The v9.8 formal bank verifies, without user axioms or admitted proofs:

- the canonical pre-square direct source as a literal source transcription, together with the fact that the physical transform $\widehat{W}_D$ is one common datum;
- the all- m generic row type, with no clean- P_3 field;
- the exact normalized and physical Gate-1A energies and their H^2 normalization bridge;
- the finite participation-to-family-energy inequality;
- the exponent conversion $R^{-1/2} \mapsto R^{-1/4}$ and the margins $1/72, 1/24, 1/32$;
- the projective/axis finite algebra and the U^{-2} error budget;
- the logical separation between the Gate-1A direct theorem and the upstream global packet/source census.

The relevant files are

<code>V98CanonicalDirectSource.lean</code>	canonical source and common-weight firewall
<code>V98CanonicalAllMRows.lean</code>	all- m row family
<code>V98DirectEnergyPin.lean</code>	exact energy objects and pinning interface
<code>V98BPPProvenance.lean</code>	external BPP participation provenance
<code>V98DirectClosure.lean</code>	deterministic normalized-to-physical compiler
<code>V98Gate0ScopeSplit.lean</code>	Gate-0/Gate-1A scope separation
<code>V98Status.lean</code>	axiom/status audit

12.2 What Lean does not prove

The formal development does *not* manufacture an inhabitant of the deep analytic BPP source input. In particular, the trigonometric Bernstein estimate, the source-specific smooth truncation, and Huxley’s short-interval prime theorem are not currently formalized in the

project. They are recorded with external/source-specific provenance, and no global axiom is introduced. The state-diagonal issue is not external: the finite moving-family bank splits diagonal and off-diagonal exactly, and the corrected projective identity places the diagonal in the $\Delta = 0$ branch.

Likewise, the formal “energy pin” is an explicit equality/interface tying a generic BPP energy to the canonical Gate-1A energy; the development includes guards showing that an arbitrary abstract bound cannot manufacture this pin. Thus a successful compiler is not confused with an independently proved analytic estimate.

12.3 Research status versus formal status

Accordingly the appropriate status language is:

research-level conditional analytic theorem under explicit external/source inputs

for the mathematics of this note, and

external-uninhabited in Lean

for the fully formal end-to-end analytic proof. The latter is a provenance boundary, not an exponent deficit or an additional arithmetic obstruction.

13 Relation to Kloosterman-fraction technology

Bettin–Chandee prove non-trivial bounds for separated trilinear Kloosterman fractions with arbitrary coefficient sequences [4]. Wright improves this architecture when a factor of the denominator is fixed [5]. These results motivated intermediate representations, but the final direct proof here does not black-box either theorem. Its decisive resource is instead smooth participation in the moving prime family.

14 What remains outside this theorem

The following are not conclusions of the main theorem:

1. exhaustive routing of every global high-complexity source packet into the canonical direct operator;
2. closure of the separate Gate-1B analytic frontier, whose present high-conductor residual is recorded in Section C;
3. Full Type-II reassembly;
4. a Hardy–Littlewood prime-pair asymptotic or twin-prime infinitude.

The point of the second item is structural. The Gate-1A theorem established here supplies one all- m analytic engine. It does not provide the signed prime-conductor character moment that remains in the complementary Gate-1B branch, and hence cannot by itself be promoted to a complete Type-II theorem.

15 Conclusion

The main conceptual point is that local sparsity alone does not give a moving-family saving. After source-preserving transport, the row coefficient remains r -dependent through a uniformly smooth envelope. Band-limited prime participation supplies the missing family coherence: smoothness creates a long plateau, prime distribution fills it with many moving rows, and finite Cauchy converts participation into the required $R^{-1/2+o(1)}$ energy contraction. After one and only one outer square root, the resulting $R^{-1/4+o(1)}$ gain closes the canonical all- m common-weight Gate-1A direct estimate throughout the parameter polytope.

A substantive feature of this mechanism is that its resource is the moving prime row r , not a prime factor of m . No largest-prime-factor quantity of m enters Theorems 6.1, 6.3 and 6.5. Thus the argument applies uniformly to the canonical all- m row family, including smooth values of m ; clean- P_3 is a positive-restriction corollary rather than the source of the saving.

A Formal theorem map and scope ledger

For ease of hostile checking, we summarize the logical status of the main components used above.

Component	Status	Role in this note
Canonical direct source and common W_D	source-pinned; finite identities Lean-checked	fixes the theorem's source class
All- m generic row family	Lean-constructed	removes clean- P_3 from the theorem statement
One-sided Poisson–Bruhat smooth envelope	explicit source/analytic input	gives $c_r(X) = B_X F_X(r/R)$
Finite BPP compiler	Lean-proved	converts participation into family energy
Huxley short-interval theorem + fixed-state exclusion	published theorem + source-specific hypothesis	supplies $P = R^{3/4-o(1)}$
Moving-family/projective diagonal partition	Lean-proved finite algebra	places $X = Y$ in the $\Delta = 0$ branch
One-root exponent ledger	Lean-proved	gives $R^{-1/4}$ and vertex margins
Projective ratio-class pushforward	Lean-proved finite algebra	controls the zero-curvature branch under the source routing pin
U^{-2} recombination budget	Lean-proved arithmetic	closes the source error
Arbitrary edge-dependent D2 packet	outside theorem scope	Gate-0 source-adaptor obligation
Global packet census / Full Type II	open downstream task	not implied by the present theorem

B Canonical source versus arbitrary edge dependence

It is useful to make the source distinction algebraically explicit. The canonical coefficient has the form

$$\omega_e(p, q, h) = R_e(p, q, h) \widehat{W}_D(h/(pq)),$$

where R_e is a deterministic scalar generated by the row and by exact reciprocity. Hence the entire family is determined once the one physical weight W_D is fixed. This is fundamentally different from prescribing an arbitrary collection

$$\{W_{D,e} : e \in \mathcal{E}^\# \}$$

with no relation between different rows. The latter can have effective rank comparable with the number of rows and therefore cannot be reduced to the canonical theorem by functional analysis alone. This is why the global source compiler must either identify a packet with the common-weight source, reduce it to finitely many smooth templates, or route it elsewhere.

C Complementary programme boundary: Gate 1B, current authoritative snapshot

This appendix is a research-status ledger, not an ingredient in Theorem 10.1. Every statement below is labelled by its current role. The controlling high-order decomposition writes, schematically,

$$a_u b_v = \prod_{i=1}^9 (f_i + \delta_i) = \sum_{J \subseteq [9]} \delta_J f_J,$$

and for an order- j term one has the determinant shell

$$C_J a_J - q\ell = -2, \quad C_J \asymp Y^j, \quad a_J \asymp Y^{9-j}.$$

Orders 1–4 are treated in the existing bank. H6 is regrouped into the same two-dimensional geometry as order 5. H7 and H8 admit one-dimensional Fourier/character representations; H9 is a separate pure-defect Type-II child.

C.1 Low/medium conductor: closed at fixed-power scale

Write the induced character modulus as

$$q = ce, \quad c \asymp C, \quad q \asymp Q.$$

After the corrected state count,

$$\sum_{c, \chi^*, e} |\beta(ce)|^2 \ll QCX^{o(1)}.$$

The coefficient-blind estimate is

$$\frac{|\mathcal{H}_s(C)|}{Y^9} \ll \begin{cases} C/Q, & C \leq Y^4, \\ C^2/(QY^4), & C \geq Y^4. \end{cases}$$

The two formulae cross at $C = Y^4$, while the zero-saving threshold is

$$C_0 = Q^{1/2}Y^2.$$

Thus the range $C \leq C_0 Y^{-\varepsilon_0}$ has a fixed-power margin and is banked at the research level. The threshold cell and larger conductors form the high-conductor problem.

C.2 Exact source split in high conductor

The squarefree source coefficient is

$$\lambda_3(U, V; q) = -\mu(q) \sum_{\substack{p|q \\ p > V \\ q/p > U}} \log p, \quad \beta(q) = \sum_{dp=q} \mu(d) \log p.$$

The physical threshold is $V = Y^{5/2-o(1)}$, while high conductor gives $e = q/c \leq Y^2$. Hence every distinguished source prime satisfies

$$p > V > e,$$

so the lane $p \mid e$ is empty. Therefore

$$c = pc_0, \quad d = c_0e,$$

and on the clean squarefree sector

$$\beta ce) = \mu(e) \rho_{C,e}(c), \quad \rho_{C,e}(c) = \sum_{pc_0=c} \mu(c_0) \log p$$

with the physical cell restrictions retained.

For a primitive character $\chi^* \pmod{c}$ induced to modulus ce ,

$$\tau_{ce}(\chi) = \mu(e) \chi^*(e) \tau_c(\chi^*).$$

Consequently

$$\beta ce) \tau_{ce}(\chi) = \rho_{C,e}(c) \chi^*(e) \tau_c(\chi^*),$$

so the short-cofactor Möbius sign is spent exactly. It is not an independent cancellation resource.

C.3 Prime-character collapse: exact

In H7 write the seven defect variables and one model variable as

$$N = r_1 \cdots r_7 x \asymp Y^8.$$

For $c = pc_0$ with $(p, c_0) = 1$, a primitive character factors as $\chi = \chi_p \chi_0$ and the Gauss sum factors as

$$\tau_{pc_0}(\chi_p \chi_0) = \chi_p(c_0) \chi_0(p) \tau_p(\chi_p) \tau_{c_0}(\chi_0).$$

The total p -local argument is

$$A_p \equiv c_0 e N (2h)^{-1} \pmod{p}.$$

For $A \in \mathbb{F}_p^\times$, orthogonality gives the exact identity

$$\sum_{\chi_p \neq \chi_{\text{pr}}} \frac{\tau_p(\chi_p)}{\sqrt{p}} \chi_p(A) = \frac{p-1}{\sqrt{p}} e_p(\bar{A}) + \frac{1}{\sqrt{p}}. \quad (20)$$

Hence the reciprocal character dependence on h becomes a linear additive prime phase

$$e_p(2h \overline{c_0 e N}).$$

The $1/\sqrt{p}$ correction is power-negligible on the clean nonprincipal c_0 sector after smooth Poisson summation.

C.4 Hybrid h -Poisson and the dual determinant

The long Fourier slot has length

$$H = Q/Y,$$

and since $c = pc_0$ and $q = ce \asymp Q$,

$$\frac{c}{H} = \frac{Y}{e}.$$

Thus the dual h -frequency window has length

$$M_h^* \asymp Y/e < p,$$

so each residue class modulo p contains at most one effective dual frequency. Fixing primitive $\chi_0 \pmod{c_0}$, the hybrid Poisson identity has the form

$$\sum_{h \in \mathbb{Z}} v(h/H) \overline{\chi_0(h)} e_p(ah) \quad (21)$$

$$= \frac{H}{c_0} \tau_{c_0}(\overline{\chi_0}) \sum_{\substack{m \in \mathbb{Z} \\ m \equiv ac_0 \pmod{p}}} \chi_0(-m\bar{p}) \widehat{v}\left(\frac{mH}{pc_0}\right), \quad (22)$$

where $a = 2\overline{c_0 e N} \pmod{p}$. The surviving frequency satisfies

$$meN \equiv 2 \pmod{p}, \quad |m| \ll (Y/e)X^{o(1)}.$$

The remaining primitive c_0 -character sum is also exact. For odd squarefree c_0 ,

$$\sum_{\chi_0(c_0)}^* \chi_0(A) = \sum_{\substack{d|c_0 \\ d|A-1}} \mu(c_0/d) \phi(d),$$

so after multiplying the source factor $\mu(c_0)/c_0$,

$$\frac{\mu(c_0)}{c_0} \sum_{\chi_0}^* \chi_0(A) = \frac{1}{c_0} \sum_{\substack{d|c_0 \\ d|A-1}} \mu(d) \phi(d). \quad (23)$$

With $A = meN/2$, the prime congruence and divisor projector combine into $pd \mid meN - 2$.

The H7 high-conductor main term is therefore reduced to the explicit dual kernel

$$\mathcal{D}_7(C) = \sum_{\substack{pc_0 \asymp C \\ p > V \\ e \asymp Q/C}} \log p \left(1 - \frac{1}{p}\right) \sum_{N \asymp Y^8} a_7(N) \sum_{\substack{|m| \ll (Y/e)X^{o(1)} \\ p \mid meN-2}} \widehat{v}(me/Y) \quad (24)$$

$$\times \frac{1}{c_0} \sum_{\substack{d|c_0 \\ d \mid meN-2}} \mu(d) \phi(d), \quad (25)$$

where $a_7(N)$ is the literal seven-defect plus one-model source coefficient. Writing $c_0 = ds$ and $n = me$, the core determinant is

$$nN - pd\ell' = 2, \quad n \ll Y, \quad N \asymp Y^8.$$

C.5 The fixed-power wall is gone; the log-saving wall remains

For fixed (e, m, N) , the integer $meN - 2$ has only $O_\eta(1)$ admissible prime divisors $p > V = Y^{5/2-o(1)}$; after p is fixed, the divisor choices are $X^{o(1)}$, and the remaining s -sum is harmonic. This yields

$$|\mathcal{D}_7(C)| \ll Y^9 X^{o(1)}. \quad (26)$$

The former coefficient-blind bound was larger by Q/Y^4 . Hence

the entire fixed-power deficit Q/Y^4 is recovered.

However the Gate target is

$$Y^9(\log X)^{-A},$$

so the current power margin is zero and arbitrary-log cancellation remains open.

There is also an exact anti-loop. The full divisor term $d = c_0$ in (25) reconstructs the original H7 two-model determinant skeleton after the dual substitution $n = me$. Thus prime-character collapse plus h -Poisson is an H7 self-duality, not by itself an iteration-to-closure. In the generic projector sector $(c_0, meN - 2) = 1$, the Möbius factor $\mu(c_0)$ is algebraically spent and leaves the positive harmonic weight $1/c_0$. The cancellation resource that remains visibly linear is the centred defect source.

Status note C.1 (Current first open: H7-DUALDET-ONEDEFECT45). The first high-order analytic open is to prove, source-specifically,

$\mathcal{D}_7(C) \ll_A Y^9(\log X)^{-A}$

while keeping at least one δ_i linear. Equivalently one may formulate a centred seven-defect determinant variance of the schematic form

$$\sum_{n \ll Y} \sum_{c \sim C} w_c \left[\sum_{\substack{N \sim Y^8 \\ nN \equiv 2(c)}} a_7(N) - \text{canonical local density} \right] \ll_A Y^9(\log X)^{-A}.$$

Status: **OPEN**. The fixed-power repair preceding this statement is **PASS**.

C.6 What remains after H7

Even a proof of the preceding H7 estimate would not by itself close Gate 1B. The controlling residual ledger is:

Component	Current status
H7 low/medium conductor	PASS
H7 prime-conductor source split, prime-character collapse, hybrid h -Poisson	PASS
H7 fixed-power repair	PASS
H7 centred dual-determinant arbitrary-log saving	OPEN
H8 high-order branch	not run to closure; OPEN
H9 pure-defect character packet	explicit Type-II child; OPEN
Order-5/6 same- q nine-factor Gram	OPEN
Historical “D12 bulk/spike + Pascadi” route	abandoned / nonclosing
Underlying cross-modulus moving- D contribution	source-specific residual; OPEN
S1/S2 literal source-normalisation pins	narrow PENDING
Canonical zero mode versus historical $E(q)$, $\mathcal{R}_E$	OPEN
Global packet census and Full Type-II re-assembly	OPEN

The abandoned D12 label should not be read as disappearance of the physical cross-modulus contribution: only that particular bulk/spike/Pascadi shortcut has been retired.

C.7 Technology map and backend audit

The programme uses several distinct tools, each in a specific role. Poisson summation, additive reciprocity, CRT, Gauss sums, primitive-character orthogonality and projective/finite-energy algebra supply exact transforms. Multiplicative and spectral large sieves provide coefficient-blind benchmarks. Bettin–Chandee type Kloosterman-fraction estimates and Wright’s partially fixed-modulus architecture motivated intermediate reductions [4, 5]. Pascadi’s non-abelian amplification gives strong fixed/composite-modulus Type-II technology [6]; it is a backend for appropriate local packets, not a black box for the current moving signed prime-conductor family. The asymptotic large sieve [11], Fouvry–Radziwiłł’s unbalanced convolution dispersion theorem [12], and the Banks–Shparlinski interval/arbitrary-set energy bounds [13] are also nearby but do not directly imply (25) with arbitrary-log saving and the seven centred defects.

C.8 Gate-1B closure DAG

The shortest current dependency picture is

$$\begin{aligned} & \text{H7 log saving} \longrightarrow \text{H8} \longrightarrow \text{H9}, \\ & \text{together with same-}q, \text{ the cross-modulus source residual, and zero-mode/source locks,} \\ & \hspace{15em} \longrightarrow \text{Full Gate-1B reassembly.} \end{aligned}$$

Only after this source-exhaustive reassembly can the result be compared with the full arbitrary-coefficient Type-II statement required downstream.

D Publication-facing theorem/status map

Object	Status	Meaning
Canonical source and determinant identities	PASS	finite/source algebra
BPP finite participation compiler	PASS	Lean-proved finite inequality
Huxley short-interval prime theorem	external published	analytic input
One-sided PB smooth envelope + negligible-state disposal	PENDING	source/analytic pin
Fixed-state exclusion dictionary	PENDING	source pin
Ratio-class projective algebra	PASS	Lean-proved finite algebra
Physical projective exceptional routing	PENDING	source pin
U^{-2} recombination exponent budget	PASS	arithmetic ledger
Literal recombination source estimate	PENDING	provenance/source pin
Gate 1A canonical theorem	CONDITIONAL	conditional on displayed pins
Gate 1B H7 fixed-power repair	PASS	research calculation
Gate 1B H7 arbitrary-log closure	OPEN	first analytic open
Gate 1B full closure	OPEN	several residuals remain
Full Type II	OPEN	not implied by 1A/1B fragments alone
Twin-prime infinitude	not claimed	downstream conclusion unavailable

References

- [1] Kevin Ford and James Maynard, *On the theory of prime producing sieves*, arXiv:2407.14368 (2024).
- [2] M. N. Huxley, *On the Difference between Consecutive Primes*, Invent. Math. 15 (1972), 164–170.
- [3] Antoni Zygmund, *Trigonometric Series*, 3rd ed., Cambridge University Press, 2002.
- [4] Sandro Bettin and Vorrapan Chandee, *Trilinear forms with Kloosterman fractions*, arXiv:1502.00769 (2015).
- [5] Thomas Wright, *Trilinear Kloosterman fractions I: partially fixed moduli and unbalanced convolutions*, arXiv:2604.25177 (2026).
- [6] Alexandru Pascadi, *Non-Abelian Amplification and Bilinear Forms with Kloosterman Sums*, Geometric and Functional Analysis (2026), doi:10.1007/s00039-026-00746-0.
- [7] Sary Drappeau, *Sums of Kloosterman sums in arithmetic progressions, and the error term in the dispersion method*, arXiv:1504.05549 (2015).
- [8] Ke Gong and Chaohua Jia, *Kloosterman sums with multiplicative coefficients*, arXiv:1401.4556 (2014).
- [9] M. A. Korolev, *On Kloosterman sums with multiplicative coefficients*, arXiv:1610.09171 (2016).
- [10] Kaisa Matomäki and Joni Teräväinen, *Products of primes in arithmetic progressions*, arXiv:2301.07679 (2023).

- [11] Brian Conrey, Henryk Iwaniec and Kannan Soundararajan, *Asymptotic Large Sieve*, arXiv:1105.1176 (2011).
- [12] Étienne Fouvry and Maksym Radziwiłł, *Level of distribution of unbalanced convolutions*, Ann. Sci. Éc. Norm. Supér. 55(2) (2022), 537–568.
- [13] William D. Banks and Igor E. Shparlinski, *Congruences with intervals and arbitrary sets*, Archiv der Mathematik 114 (2020), 1–13.